MATRIX | INDUSTRIAL MAINTENANCE

Troubleshooting & Fault-Finding Closed Loop Industrial Control Systems - Answers

MATRIX

CP6773

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INTRODUCTION

This document highlights all the answers to the questions posed in the student handout for the worksheets curriculum “CP6773 - Troubleshooting & Fault-Finding Closed Loop Systems”.

This is the second curriculum, please complete the **CP0539 - Industrial Maintenance of Closed Loop Systems** curriculum first before starting this **Fault Scenarios** one.

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ANSWERS

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Answers

Scenario Solutions

The following explains the correct solution for each of the fault scenarios.

Scenario 1: Tank Temperature Too High

Solution: The tank offset was incorrectly left on last night, adding 33 degrees, the student needs to highlight this and set the offset back to zero to successfully pass this scenario.

Scenario 2: Tank Lid Flicker

Solution: The proximity switch on the tank lid keeps flicking on and off every 5 seconds. The student must notice this on the IO screen.

Scenario 3: Bad Flow

Solution: The student must notice on the flow manometer gauge that the system is delivering approximately 50 litres a minute too much flow. This is due to an offset added to the flow sensor on the IO screen. They should remove this.

Scenario 4: Tank Temperature Too Low

Solution: The tank temperature is blipping to below 0 degrees every 15 seconds or so. The fault only remains for a few seconds before clearing.

Scenario 5: High Float Switch

Solution: Correctly deduce the tanks high level float switch is the problem. There’s no signal for it on the IO screen.

Scenario 6: Estop Channel

Solution: Correctly deduce one of the Estop channels is the problem.

Scenario 7: Temperature Scalar

Solution: Correctly deduce the temperature sensor has been scaled incorrectly.

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Scenario 8: Settings

Solution: Highlight that PID manual mode is on and that the Kp values are different from their original settings highlighted on that Setup page. The student should reset the PID parameters using the Reset PID button.

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Fault Scenario Explanations

Should you wish to explain possible reasons for each fault scenarios, you can do so using the below explanations;

Scenario 1 Explanation: Tank Temperature Too High

A mistake like this is common in industry because operators sometimes add temporary offsets to compensate for sensor errors or delays in heating. If they forget to remove the offset after troubleshooting, the system continues to run with incorrect temperature readings, causing overheating or process failures.

Scenario 2 Explanation: Tank Lid Flicker

An issue like this would happen in industry because proximity switches are prone to misalignment over time due to vibrations, repeated use, or mechanical wear. If the sensor isn’t properly checked during maintenance, it may flicker on and off, falsely indicating that the lid is open and preventing the system from running.

Scenario 3 Explanation: Bad Flow

This kind of problem tends to occur in industry because flow issues can be caused by blockages, airlocks in the piping, pump cavitation, or partially closed valves. Wear and tear on pumps or leaks in the system can also reduce flow, while buildup of debris in filters or strainers can create unexpected restrictions. If these issues aren't identified and fixed, the system struggles to maintain proper flow rates, leading to unstable operation.

Scenario 4 Explanation: Tank Temperature Too Low

A situation like this can arise in industry because maintenance teams often apply offsets to sensor readings when trying to match expected values or troubleshoot a faulty sensor. If the offset isn’t removed afterward, the system continues operating with incorrect temperature data, leading to cold tanks and process failures.

Scenario 5 Explanation: High Float Switch

This kind of failure is common in industry because float switches often degrade over time due to wear, contamination, or electrical interference. If a switch starts triggering

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intermittently due to mechanical sticking or water ingress, it can falsely indicate high levels, leading to unnecessary shutdowns.

Scenario 6 Explanation: Estop Channel

This would often happen in industry because emergency stop circuits can sometimes experience frequent use, leading to worn-out contacts, broken wiring, or corrosion. If an Estop channel begins failing, the system may incorrectly detect a fault, causing unexpected shutdowns and requiring troubleshooting. In this scenario it was mentioned that a repair was done. These can sometimes not be done correctly.

Scenario 7 Explanation: Temperature Scalar

This is a common problem in industry because incorrect scaling can be applied to sensors during installation or recalibration to match other equipment. If the scaling isn't verified properly, the system operates with skewed temperature data, affecting performance and triggering unnecessary alarms.

Scenario 8 Explanation: Settings

In industry, PID values are sometimes shown on the HMI for engineers to adjust. This can be dangerous but none the less, inexperienced people do tend to change these from time to time. They often change the Kp parameter hoping to make the system go "faster", not realising the system was set this way for a reason. Often times there is no "reset" button to roll the PID parameters back. In this case the system will have to be retuned or recommissioned.

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VERSION CONTROL

15 05 25	First Revision Created

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